

55. Inverse Laplace transform of $\frac{s+1}{s^2-4}$ is

- (a) $\cos 2x + \frac{1}{2} \sin 2x$ (b) $\cos x + \frac{1}{2} \sin x$
 (c) $\cosh x + \frac{1}{2} \sinh x$ (d) $\cosh 2x + \frac{1}{2} \sinh 2x$

56. The points, where the series solution of the Legendre differential equation

$$(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + \frac{3}{2} \left(\frac{3}{2} + 1 \right) y = 0$$

will diverge, are located at

- (a) 0 and 1 (b) 0 and -1
 (c) -1 and 1 (d) $\frac{3}{2}$ and $\frac{5}{2}$

57. Solution of the differential equation $x \frac{dy}{dx} + y = x^4$,

with the boundary condition that $y = 1$, at $x = 1$, is

- (a) $y = 5x^4 - 4$ (b) $y = \frac{x^4}{5} + \frac{4x}{5}$
 (c) $y = \frac{4x^4}{5} + \frac{1}{5x}$ (d) $y = \frac{x^4}{5} + \frac{4}{5x}$

58. Can the following scalar and vector potentials describe an electromagnetic field? $\phi(\vec{x}, t) = 3xyz - 4t$

$$\vec{A}(\vec{x}, t) = (2x - \omega t) \hat{i} + (y - 2z) \hat{j} + (z - 2e^{i\omega t}) \hat{k}$$

where, ω is a constant.

- (a) Yes, in the Coulomb gauge
 (b) Yes, in the Lorentz gauge
 (c) Yes, provided $\omega = 0$
 (d) No

GATE 2006

59. A linear transformation T , defined as $T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 - x_3 \end{bmatrix}$,

transforms a vector $\vec{x}$ for a three-dimensional real space to a two-dimensional real space. The transformation matrix T is

- (a) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$
 (c) $\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

60. The value of $\oint_S \frac{\vec{r} \cdot d\vec{S}}{r^3}$, where $\vec{r}$ is the position vector

and S is a closed surface enclosing the origin, is

- (a) zero (b) π
 (c) 4π (d) 8π

61. The value of $\oint_C \frac{e^{2z}}{(z+1)^4} dz$, where C is a circle defined by $|z| = 3$, is

- (a) $\frac{8\pi i}{3} e^{-2}$ (b) $\frac{8\pi i}{3} e^{-1}$
 (c) $\frac{8\pi i}{3} e$ (d) $\frac{8\pi i}{3} e^2$

62. The k th Fourier component of $f(x) = \delta(x)$ is

- (a) 1 (b) zero
 (c) $(2\pi)^{-1/2}$ (d) $(2\pi)^{-3/2}$

Common Data for Questions 63 and 64

One of the eigen values of the matrix $\begin{bmatrix} 2 & 3 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is 5.

63. The other two eigen values are

- (a) 0 and 0 (b) 1 and 1
 (c) 1 and -1 (d) -1 and -1

64. The normalized eigen vector corresponding to the eigen value 5 is

- (a) $\frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$ (b) $\frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$
 (c) $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ (d) $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

GATE 2005

65. If a vector field $\vec{F} = x\hat{i} + 2y\hat{j} + 3z\hat{k}$, then $\vec{\nabla} \times (\vec{\nabla} \times \vec{F})$ is

- (a) zero (b) $\hat{i}$
 (c) $2\hat{j}$ (d) $3\hat{k}$

66. All solutions of the equation $e^z = -3$ are

- (a) $z = i n \pi \ln 3, n = \pm 1, \pm 2, \dots$
 (b) $z = \ln 3 + i (2n + 1) \pi, n = 0, \pm 1, \pm 2, \dots$
 (c) $z = \ln 3 + i 2 n \pi, n = 0, \pm 1, \pm 2, \dots$
 (d) $z = i 3 n \pi, n = \pm 1 \pm 2, \dots$

67. If $\bar{f}(s)$ is the Laplace transform of $f(t)$ the Laplace transform of $f(at)$, where a is a constant is

- (a) $\frac{1}{a} \bar{f}(s)$ (b) $\frac{1}{a} \bar{f}\left(\frac{s}{a}\right)$
 (c) $\bar{f}(s)$ (d) $\bar{f}\left(\frac{s}{a}\right)$

68. Given the four vectors

$$u_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, u_2 = \begin{bmatrix} 3 \\ -5 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} 2 \\ 4 \\ -8 \end{bmatrix}, u_4 = \begin{bmatrix} 3 \\ 6 \\ -12 \end{bmatrix}$$

The linearly dependent pair is

- (a) u_1, u_2 (b) u_1, u_3
 (c) u_1, u_4 (d) u_3, u_4

69. Which of the following functions of the complex variable z is not analytic everywhere?

- (a) e^z (b) $\sin \frac{z}{z}$
 (c) e^3 (d) $|z|^3$

70. Eigen values of the matrix

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2i \\ 0 & 0 & 2i & 0 \end{bmatrix}$$

are

- (a) $-2, -1, 1, 2$ (b) $-1, 1, 0, 2$
(c) $1, 0, 2, 3$ (d) $-1, 1, 0, 3$

Statement for Linked Answer Questions 71 and 72

For the differential equation $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0$

71. One of the solution is
(a) e^x (b) $\ln x$
(c) e^{-x^2} (d) e^{-x}
72. The second linearly independent solution is
(a) e^{-x} (b) xe^x
(c) $x^2 e^x$ (d) $x^2 e^{-x}$

GATE 2004

73. A real traceless 4×4 matrix has two eigen values -1 and $+1$. The other eigen values are

- (a) zero and $+2$ (b) -1 and $+1$
(c) zero and $+1$ (d) $+1$ and $+1$

74. The eigen values of the matrix $\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$ are

- (a) $+1$ and $+1$ (b) zero and $+1$
(c) zero and $+2$ (d) -1 and $+1$

75. The inverse of the complex number $\frac{3+4i}{3-4i}$ is

- (a) $\frac{7}{25} + i\frac{24}{25}$ (b) $-\frac{7}{25} + i\frac{24}{25}$
(c) $\frac{7}{25} - i\frac{24}{25}$ (d) $-\frac{7}{25} - i\frac{24}{25}$

76. The value of $\int_C \frac{dz}{(z^2 + a^2)}$, where C is a unit circle

(anticlockwise) centred at the origin in the complex Z -plane is

- (a) π for $a = 2$ (b) zero for $a = \frac{1}{2}$
(c) 4π for $a = 2$ (d) $\frac{\pi}{2}$ for $a = \frac{1}{2}$

77. The Laplace transform of $f(t) = \sin \pi t$ is $F(s) = \frac{\pi}{s^2(s^2 + \pi^2)}$

$s > 0$. Therefore, the Laplace transform of $t \sin \pi t$ is

- (a) $\frac{\pi}{s^2(s^2 + \pi^2)}$ (b) $\frac{2\pi}{s^2(s^2 + \pi^2)^2}$
(c) $\frac{2\pi s}{(s^2 + \pi^2)^2}$ (d) $\frac{2\pi}{(s^2 + \pi^2)^2}$

78. A periodic function $f(x) = x$ for $-\pi < x < +\pi$ has the Fourier series representation

$$f(x) = \sum_{n=1}^{\infty} \left(-\frac{2}{n}\right) (-1)^n \sin nx$$

Using this, one finds the sum $\sum_{n=1}^{\infty} n^{-2}$ to be

- (a) $2 \ln 2$ (b) $\frac{\pi^2}{3}$
(c) $\frac{\pi^2}{6}$ (d) $\pi \ln 2$

79. The Fourier transform $F(k)$ of a function $f(x)$ is defined as

$$F(k) = \int_{-\infty}^{\infty} dx f(x) \exp(ikx). \text{ Then } F(k) \text{ for}$$

$f(x) = \exp(-x^2)$ is [Given: $\int_{-\infty}^{\infty} \exp(-x^2) dx = \sqrt{\pi}$]

- (a) $\pi \exp(-k)$ (b) $\sqrt{\pi} \exp\left(-\frac{k^2}{4}\right)$
(c) $\frac{\sqrt{\pi}}{2} \exp\left(-\frac{k^2}{2}\right)$ (d) $\sqrt{2\pi} \exp(-k^2)$

80. The Lagrangian of a particle moving in a plane under the influence of a central potential is given by

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r). \text{ The generalized momenta}$$

corresponding to r and θ are given by

- (a) $m\dot{r}$ and $m r^2 \dot{\theta}$ (b) $m\dot{r}$ and $m r \dot{\theta}$
(c) $m r^2$ and $m r^2 \dot{\theta}$ (d) $m r^2$ and $m r^2 \dot{\theta}^2$

Common Data for Questions 81 and 82

Consider the differential equation $y' + p(x)y' + q(x)y(x) = 0$.

81. If $x p(x)$ and $x^2 q(x)$ have the Taylor series expansions

$$x p(x) = 4 + x + x^2 + \dots$$

$$x^2 q(x) = 2 + 3x + 5x^2 + \dots$$

Then the roots of the indicial equation are

- (a) $-1, 0$ (b) $-1, -2$
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82. If $p(x) = 0$ with the Wronskian at $x = 0$ as $W(x = 0) = 1$ and one of the solutions is x , then the other linearly independent solution which vanishes at $x = 1/2$ is

- (a) 1 (b) $1 - 4x^2$
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GATE 2003

83. The curl of the vector $\vec{A} = z\hat{i} + x\hat{j} + y\hat{k}$ is given by

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84. Consider the differential equation

$$\frac{d^2 x}{dt^2} + 2 \frac{dx}{dt} + x = 0$$

At time $t = 0$, it is given that $x = 1$ and $\frac{dx}{dt} = 0$. At $t = 1$, the value of x is given by

6 Chapterwise GATE Solved Papers

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GATE 2005

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(c) 2 (d) 3k

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$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 2 & 0 \end{bmatrix}$$

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